[This question paper contains 8 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 4717 **E**

Unique Paper Code : 32517916

Name of the Paper : Embedded Systems

Name of the Course : **B.Sc. (H) Electronics (DSE)**

Semester : VI (CBCS)

Duration : 3 Hours Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. There are **seven** questions in all, out of which you have to attempt any **Five** questions.
3. Q1. is compulsory. **All** Questions carry equal marks.
4. Use of non-programmable scientific calculator is allowed.

1. (a) State any three key differences between a microcontroller and a general-purpose microprocessor. (3)

P.T.O.

(b) Give two features of GPR and SFR in AVR. (3)

(c) What is the function of SP register and the size of SP register in Atmega32? (3)

(d) Calculate the overhead percentage if the data size is 7, 1 stop bit, 1 start bit and no parity bit. (3)

(e) Find the step size for an 8 bit ADC; if V_{REF} = 1.28 V. Also calculate the D_0-D_7 output if the analog input is 1 V. (3)

2. (a) Explain and compare Von Neumann and Harvard architecture in terms of arrangement, speed, space and hardware requirements. (5)

(b) Show the status of the flags C, H, and N flags and values of registers R16, and R17 after the execution of each line of the assembly code shown below *[Assuming initial conditions, C=0, H=0, N=0, R16=0x00, R17=0x00]*:

AVR PSC


```
.INCLUDE "M32DEF.INC"
LDI R16, 35
LDI R17, 99
ADD R16, R17
LDI R17, 0xE5
SUB R16, R17
```

(5)

(c) Differentiate between

(i) SBCI and SBIC

(ii) LSL and LSR

Use examples to explain. (5)

3. (a) What is a Program counter? What are the highest and lowest values in hex that the ATmega32 program counter can take? How its value is affected by the CALL and RET commands? (5)

(b) Write the addressing modes used in the following instructions : (5)

```
COM R20
MUL R16, R20
STS 0x67, R21
IN R24, PINB
ST X, R20
```

(e) Which of the following instructions are legal or illegal. Also, give reasons why the instructions are illegal (5)

```
LDI R14, $45
ADD R32, R16
LDI R17, $105
IN R1, $10
STS PINC, R2
MOV R10, R20
```

4. (a) What are the maximum and minimum number of machine cycles needed to execute an AVR instruction? Give any two instructions which require maximum number of machine cycles.

Assuming a crystal frequency of 10 MHz, find the time delay associated with the loop section of the following DELAY subroutine (5)

```
DELAY:   LDI R20, 0X64
HERE:    NOP
         NOP
         NOP
         NOP
         DEC R20
         BRNE HERE
         RET
```

(b) Briefly explain the data memory map of Atmega32. (5)

(c) Write a program to convert ASCII digits of '3' and '6' to packed BCD and display on Port D. (5)

5. (a) List the various clock sources in AVR Atmega32. What is the default clock source? Which clock source do you suggest if you need (5)

(i) A variable clock source

(ii) To build a system with minimum external hardware Which reset mechanism is used by ATMega microcontroller family to provide safety against power source fluctuations?

(b) Upon power-on reset of the Atmega32, what memory area is assigned to the interrupt vector table? Can the programmer change the memory space assigned in the table?

Show the instructions needed to enable the Timer0 compare match interrupt. (5)

(c) Assume that Timer0 is programmed in Normal mode, TCNT0 = 0X50 and the Timer0 overflow interrupt bit is enabled. Explain how the interrupt for the timer works. Assume XTAL = 8 MHz with prescaler = 256, How often the ISR is executed? (5)

6. (a) Using polling and INTI, write a program that toggles bit PORTB.5 on every rising edge. (5)

(b) Assume that XTAL = 8 MHz. Find the TCNTO value needed to generate a time delay of 5 ms. Use Normal mode and the largest prescaler possible. Write the instructions to set the Timer0 in this mode. (5)

(c) Write a program for the AVR to transmit the letter 'X' serially at 4800 baud, continuously. Use interrupts instead of the polling method. Assume XTAL = 8MHz, 8 bit data, odd parity and 2 stop bits. (5)

7. (a) Write a program to receive analog input voltage from ADC7 (ADC channel 7) and display 10-bit digital data output at PORTB and PORTD. (Higher byte is displayed at PORTB and lower byte at PORTD). Select Clk/64, V_{REF} = 2.56 V and left justified data. (5)

(b) Explain the Address packet format and data packet format in I2C. Also, explain what is clock stretching? (5)

(c) Explain the various steps needed for writing data to an SPI device? State the difference between the single-byte and burst modes in terms of the CE signal. (5)

Appendix

GICR	INT1	INT0	INT2				IVSEL	IVCE
GIFR	INTF1	INTF0	INTF2					
TIMSK	OCIE2	TOIE2	TICIE1	OCIE1A	OCIE1B	TOIE1	OCIE0	TOIE0
TIFR	OCF2	TOV2	ICF1	OCF1A	OCF1B	TOV1	OCF0	TOV0
UCSRA	RXC	TXC	UDRE	FE	DOR	PE	U2X	MPCM
UCSRB	RXCIE	TXCIE	UDRIE	RXEN	TXEN	UCSZ2	RXB8	TXB8
UCSRC	URSEL	UMSEL	UPM1	UPM0	USBS	UCSZ1	UCSZ0	UCPOL
TCCR0	FOC0	WGM00	COM01	COM00	WGM01	CS02	CS01	CS00
MCUCR	SE	SM2	SM1	SM0	ISC11	ISC10	ISC01	ISC00
ADCSRA	ADEN	ADSC	ADATE	ADIF	ADIE	ADPS2	ADPS1	ADPS0
ADMUX	REFS1	REFS0	ADLAR	MUX4	MUX3	MUX2	MUX1	MUX0